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METHODS ARTICLE

Evaluating AI-Based Psychological Interventions

Designs, control conditions, outcomes, safety monitoring and reporting standards

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ABSTRACT

Conversational agents and other AI-based tools are increasingly offered as psychological interventions, yet the methods used to evaluate them have not kept pace with their technical development. This methodological contribution outlines the main design decisions that determine whether a study of an AI-based intervention can support credible conclusions. We first locate evaluation questions along a development cycle, from feasibility and optimization studies, including factorial, adaptive and micro-randomized designs, to confirmatory randomized controlled trials. We then discuss the choice of control conditions, arguing that waitlist and information-only comparators answer narrower questions than is often assumed and that attention-matched or active comparators are needed for claims about specific benefits. Outcome selection should combine validated symptom measures with indicators of functioning, engagement and attrition, and follow-up should extend beyond the immediate post-treatment assessment. Generative systems add distinctive requirements: prospective safety monitoring with predefined adverse events, procedures for detecting harmful or inaccurate outputs, and documentation of the model version and configuration used throughout a trial. The SPIRIT-AI and CONSORT-AI extensions provide a reporting framework for these elements. We conclude that methodological rigor is a precondition for any responsible clinical use of AI in psychotherapy, including meaning-centered approaches.

Keywords: artificial intelligence · randomized controlled trials · control conditions · safety monitoring · CONSORT-AI

1 Introduction

Within a decade, AI-based psychological interventions have moved from rule-based conversational agents delivering scripted cognitive behavioral content to systems built on large language models that generate open-ended replies (Abd-Alrazaq et al., 2020; Torous et al., 2021). Early randomized trials, such as the evaluation of Woebot with young adults (Fitzpatrick et al., 2017), and more recent trials of generative chatbots (Heinz et al., 2025) have reported encouraging symptom changes. At the same time, systematic reviews have repeatedly described the evidence base as heterogeneous, short-term and of limited methodological quality (Abd-Alrazaq et al., 2020).

This article does not review effectiveness; that task is taken up elsewhere in this issue. Its aim is methodological: to describe the decisions about design, comparators, outcomes, safety monitoring and reporting that determine what a study of an AI-based intervention can and cannot show. The argument draws on established principles of psychotherapy research and on the reporting extensions developed for trials of AI interventions (Cruz Rivera et al., 2020; Liu et al., 2020). Throughout, the guiding question is a modest one: what would a skeptical but fair reader need to know before trusting a reported benefit?

2 Matching Designs to the Development Cycle

Not every study needs to be a confirmatory randomized controlled trial (RCT), but every study should state which question it addresses. Mohr et al. (2017) argued that digital mental health research has suffered from evaluating products that change faster than trials can be completed, and from neglecting the implementation contexts in which tools will actually be used. A useful response is to distinguish phases: feasibility and acceptability studies, optimization studies that identify active components, and confirmatory trials that test a fixed, well-specified intervention against an appropriate comparator.

For the optimization phase, the multiphase optimization strategy (MOST) and sequential multiple assignment randomized trials (SMART) allow investigators to examine components and adaptive decision rules before committing to a full package (Collins et al., 2007). Micro-randomized trials, in which prompts or messages are randomized repeatedly within each participant, are suited to questions about the momentary effects of specific chatbot actions, such as whether a reflective prompt sent after a reported low mood is followed by better coping in the next hours (Klasnja et al., 2015). Such designs fit conversational agents well, because individual turns can be logged and varied systematically. Their results, however, inform the design of an intervention; they do not by themselves establish its clinical effectiveness.

3 Control Conditions and Comparators

The choice of comparator defines the causal question of a trial. Mohr et al. (2009) showed for psychological interventions in general that control conditions differ in how far they control for expectancy, attention, time and contact, and that these differences shape the size and the meaning of observed effects. A waitlist control, as used in the trial of a generative chatbot reported by Heinz et al. (2025), answers whether offering the intervention is better than offering nothing for a limited period. An information-only control, as in the early Woebot trial (Fitzpatrick et al., 2017), adds a minimal credible activity but does not match the interactive, conversational character of the intervention.

Claims that a conversational agent has specific therapeutic effects therefore require stronger comparators. Options include attention-matched controls, such as a conversational agent that engages users without delivering the hypothesized active content; established digital interventions, such as guided or unguided internet-delivered cognitive behavioral therapy; and, where the question is substitution, human-delivered care. Each option carries trade-offs in feasibility, blinding and ethical acceptability. Because participants usually know whether they are interacting with a chatbot, expectancy should at least be measured at baseline, and the credibility of the control condition should be assessed rather than assumed. Non-inferiority designs, which are often invoked in this context, require a prespecified margin and an adequately powered sample.

4 Outcomes, Engagement and Follow-Up

Primary outcomes should be validated, clinically meaningful measures specified in advance, ideally complemented by assessments of functioning and quality of life. For meaning-centered approaches, measures of meaning in life can be added as secondary outcomes, provided that their inclusion is justified theoretically rather than by mere availability. Outcomes collected only within the application itself risk conflating measurement with intervention, because the same conversational context that delivers the intervention also elicits the self-report. Where clinician-rated outcomes are feasible, assessors should be blind to allocation, and remote assessment procedures should be described.

Engagement and attrition require particular attention. Eysenbach (2005) described a “law of attrition” in eHealth trials, and a meta-analytic review of smartphone-delivered interventions reported substantial dropout across studies (Linardon & Fuller-Tyszkiewicz, 2020). Analyses should follow the intention-to-treat principle, handle missing data with methods appropriate to the assumed missingness mechanism, and report engagement in detail, for example the number of sessions, messages and active days. Engagement should not be treated as a proxy for benefit: heavy use may reflect helpfulness, but also dependency or unresolved distress. Finally, follow-up periods of a few weeks, which remain common, cannot show whether effects persist; designs should include assessments several months after the end of the active phase.

5 Safety Monitoring and Adverse Events

Psychotherapy research has long recognized that interventions can cause harm, and consensus work on internet interventions recommends that negative effects be defined, monitored and reported systematically (Rozental et al., 2014). For AI-based interventions this obligation becomes more concrete. Trial protocols should specify which events count as adverse, for instance symptom deterioration, emerging suicidal ideation, crisis disclosures that are not handled appropriately, or responses that reinforce harmful behavior, and should describe how such events are detected, escalated and documented.

Generative systems introduce an additional source of risk: they can produce fluent but incorrect or unsuitable content, a phenomenon discussed in the computational literature as hallucination (Ji et al., 2023). Safety monitoring should therefore combine

participant-reported adverse events with structured review of conversation transcripts, for example clinician audits of a random sample of conversations and of all conversations flagged by automated classifiers. Heinz et al. (2025) described human oversight of chatbot responses during their trial; such procedures, and how often reviewers had to intervene, belong in the trial report. Stade et al. (2024) have proposed a staged development and evaluation of large language model applications in behavioral healthcare, in which autonomy is extended only as safety evidence accumulates.

6 Model Versioning, Transparency and Reporting

A conventional psychological intervention can be described in a manual; a system based on a large language model is defined by the underlying model, its version, the system prompt, retrieval sources, safety filters and decoding settings. Any of these may change during a trial, sometimes beyond the investigators' control when a commercial model is accessed through an interface maintained by a provider. Protocols should therefore freeze or document the configuration, record every change with its date, and plan sensitivity analyses for periods with different versions. Without this information, a trial result cannot be attributed to a reproducible intervention. Prompt texts and safety rules are part of the intervention and should be made available, at least to reviewers, in the same way as treatment manuals.

The SPIRIT-AI and CONSORT-AI extensions address these issues for protocols and reports (Cruz Rivera et al., 2020; Liu et al., 2020). Among other things, they ask investigators to state the version of the AI system, to describe how input data were acquired and how poor-quality or unavailable inputs were handled, to specify the human–AI interaction and the output of the system, and to report any analysis of performance errors. Although these extensions were developed with diagnostic and decision-support applications in mind, their logic transfers to conversational interventions. Preregistration, a publicly available protocol and, where legally and ethically possible, the sharing of de-identified conversation samples would further strengthen trust.

7 Conclusion

Evaluating AI-based psychological interventions requires no new epistemology, but it does require applying familiar principles with particular care. The question addressed should be matched to the design; comparators should be strong enough to support the claims made; outcomes should extend beyond symptoms and beyond the first weeks; safety should be monitored prospectively and transparently; and the evaluated system should be documented precisely enough to be identified again. None of these requirements is exotic; most are already standard in trials of psychological and pharmacological treatments.

From a meaning-centered perspective, methodological rigor also has an ethical dimension. Persons who turn to a digital tool in distress entrust it with part of their freedom and responsibility, and they deserve to know whether the help they are offered has been tested honestly. Careful evaluation is therefore not an obstacle to innovation but a condition for using it responsibly.

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